MODULE 23 · DOMAIN 1 - AGENTS AND WORKFLOWS

Workflows or Agents?

Predefined code paths or model-directed loops: the augmented LLM, five workflow patterns, when autonomy earns its cost, and how agents stop.

DOMAIN 1 of 8 EXAM WEIGHT 14.7% MODULE 23 of 36

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A train follows a timetable. A taxi driver picks the route. **Use the taxi only when nobody can write the timetable.**

Workflow or agent?

WORKFLOW

Predefined code paths

- Your code fixes the steps and their order
- Claude does the work inside each step
- Predictable, testable, cheaper
- For tasks you can decompose in advance

AGENT

Model-directed

- Claude decides its own steps and tool calls
- Loops until done or a limit is hit
- Flexible, but costlier and less predictable
- For open-ended tasks with unknown steps

ON THE EXAM

Anthropic: workflows use "predefined code paths"; agents "dynamically direct their own processes and tool usage".

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The augmented LLM

- **The building block:** "an LLM enhanced with augmentations such as retrieval, tools, and memory".
- **Often enough on its own:** one well-prompted call with the right tools and context.
- **Workflows arrange several** of these blocks in code.
- **Agents give one** of them a loop and the freedom to choose.
- **Start here;** add structure or autonomy only when evals show you must.

> THE TIEBREAKER
>
> Anthropic: find "the simplest solution possible, and only increasing complexity when needed".

Five workflow patterns

PATTERN	USE IT WHEN
Prompt chaining	The task decomposes cleanly into fixed subtasks
Routing	Distinct categories are better handled separately
Parallelisation	Subtasks are independent (sectioning), or you want confidence (voting)
Orchestrator-workers	You cannot predict the subtasks in advance
Evaluator-optimiser	There are clear evaluation criteria to draft against

ON THE EXAM

Match the pattern to the shape of the task. Most real systems combine two or three.

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Copperlet's repair triage: a workflow

01
Route
Emergency keywords go straight to on-call.

02
Chain
Extract, check, classify: module 10's stages.

03
Parallel
Draft reply and agency note at the same time.

04
Evaluate
A grader checks the draft against the rubric.

05
Gate
An agent approves before anything is sent.

THE TAKEAWAY

Every step is known in advance, so triage stays a workflow. No agent loop is needed.

When an agent earns its cost

- **Open-ended:** "difficult or impossible to predict the required number of steps".
- **The path depends on what it finds:** each result decides the next action.
- **A trusted environment:** tools are scoped and mistakes are recoverable.
- **Success can be checked:** tests pass, a record reconciles, a reviewer approves.
- **The price:** more tokens, more latency, compounding errors. Test in sandboxes, with guardrails.

> THE TIEBREAKER
>
> If you can draw the flowchart in advance, build the flowchart. If you cannot, consider an agent.

How agents stop

- **Natural end:** the model finishes, stop_reason end_turn.
- **Hard limits in code:** a maximum number of turns, a token or spend budget, a wall-clock timeout.
- **Checkpoints:** pause for human approval before risky or irreversible steps.
- **Retry budgets** per tool, so one failing call cannot loop forever.
- **A designed end state:** on a limit, return progress and a clear status, not silence.

> ON THE EXAM
>
> Anthropic: it is "common to include stopping conditions (such as a maximum number of iterations)".

Copperlet's agency-ops agent: an agent

WHY AN AGENT

Unknown steps

- Each arrears case is different
- Next lookup depends on what was found
- CRM, payments, messages, email
- Success: a summary an agent can verify

HOW IT IS BOUNDED

Limits by design

- Read-only tools; drafts only
- 25 turns, a token budget, 10 minutes
- Per-tool retry budget
- On a limit: partial findings, flagged

THE TAKEAWAY

The same company runs both: triage as a workflow, investigations as an agent. The task decides.

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> A company wants to translate product descriptions into French, then check each translation for brand terms, then format it for the website. The steps never change.
>
> The team is considering an autonomous agent with all three abilities as tools.

What should they build?

A. An autonomous agent that decides which step to do next

B. A multi-agent system with one agent per step

C. A prompt chain: translate, check brand terms, format, with a gate between steps

D. A single prompt asking for all three steps, with no checks

◆ PAUSE AND DECIDE

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What should they build?

A An autonomous agent that decides which step to do next

Autonomy for steps that never change.

B A multi-agent system with one agent per step

Even more complexity, same fixed path.

C A prompt chain: translate, check brand terms, format, with a gate between steps

Fixed subtasks: chaining.

D A single prompt asking for all three steps, with no checks

Loses the brand-term gate.

THE SIGNAL

"The steps never change": a workflow. Prompt chaining fits fixed, ordered subtasks.

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A support inbox receives billing questions, technical faults and cancellation requests. Each type needs different instructions, tools and a different model size. One general prompt handles all three, and quality is uneven.

The categories are stable and easy to tell apart.

Which pattern fits best?

A Evaluator-optimiser, so every answer is critiqued and redrafted

B An autonomous agent that decides how to treat each message

C Voting, running each message through the general prompt three times

D Routing: classify each message, then send it to a specialised path

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Which pattern fits best?

A Evaluator-optimiser, so every answer is critiqued and redrafted

Polishes the same general answer.

B An autonomous agent that decides how to treat each message

Autonomy where categories are known.

C Voting, running each message through the general prompt three times

Three copies of the same weakness.

D Routing: classify each message, then send it to a specialised path

Distinct categories, separate handling.

THE SIGNAL

Distinct, stable categories needing different handling: routing.

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> A team must review 40-page contracts for five independent risks: liability, termination, data protection, payment terms and governing law. One prompt checking all five is slow and sometimes misses risks.
>
> Each check can be done on its own.

Which pattern fits best?

A Sectioning: run the five checks in parallel, then merge

B An agent that decides which risks to look for in each contract

C A longer single prompt that lists all five risks in more detail

D Routing each contract to one risk check, chosen by contract type

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Which pattern fits best?

A Sectioning: run the five checks in parallel, then merge

Focused checks, at the same time.

B An agent that decides which risks to look for in each contract

The five risks are already known.

C A longer single prompt that lists all five risks in more detail

Still one overloaded, slow call.

D Routing each contract to one risk check, chosen by contract type

Every contract needs all five.

THE SIGNAL

Independent subtasks plus speed: sectioning. Each check gets full attention.

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> An engineering tool must make a requested change across a codebase. Which files need editing, and how many, is only clear after inspecting the code. A lead model can plan and assign the edits.
>
> Each edit can be made and checked separately.

Which pattern fits best?

A. A fixed prompt chain that edits a predefined list of files

B. Evaluator-optimiser on a single file

C. Orchestrator-workers: a lead model plans and delegates the edits

D. Routing to one of several fixed edit templates

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Which pattern fits best?

A. A fixed prompt chain that edits a predefined list of files
The list is not known in advance.

B. Evaluator-optimiser on a single file
The change spans many files.

C. Orchestrator-workers: a lead model plans and delegates the edits
Unpredictable subtasks, delegated.

D. Routing to one of several fixed edit templates
Assumes a fixed set of changes.

THE SIGNAL

"Only clear after inspecting" means the subtasks cannot be predicted: orchestrator-workers.

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A legal-tech tool drafts letters that must meet a detailed checklist: correct parties, statutory wording, deadlines and tone. First drafts often miss one or two items. The checklist is precise and can be graded.

Quality matters more than speed.

Which pattern fits best?

A Routing letters by type to different prompts

B Evaluator-optimiser: draft, grade, revise, within a round limit

C Parallelisation, drafting five letters and sending the longest

D An autonomous agent with web search

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Which pattern fits best?

A Routing letters by type to different prompts

Does not catch the missed items.

B Evaluator-optimiser: draft, grade, revise, within a round limit

Clear criteria, iterative fixing.

C Parallelisation, drafting five letters and sending the longest

Longest is not most correct.

D An autonomous agent with web search

Adds reach, not checking.

THE SIGNAL

Clear, gradable criteria and drafts that miss items: evaluator-optimiser, with a round limit.

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A fraud team wants Claude to investigate suspicious accounts. Each case needs a different sequence of lookups across five systems, depending on what earlier lookups reveal. Investigators verify every finding before acting.

The tools are read-only.

What should the team build?

A. A fixed prompt chain that queries all five systems in the same order every time

B. A single prompt with all five systems' data pasted in

C. Routing each case to one system, based on account type

D. A bounded agent with read-only tools and investigator review

◆ PAUSE AND DECIDE

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What should the team build?

A. A fixed prompt chain that queries all five systems in the same order every time
The order depends on findings.

B. A single prompt with all five systems' data pasted in
Bloated, and misses follow-up lookups.

C. Routing each case to one system, based on account type
Cases need several systems.

D. A bounded agent with read-only tools and investigator review
Unknown steps, bounded safely.

THE SIGNAL

The next step depends on what was found, tools are safe, results are checked: an agent earns its cost.

> An agent that reconciles invoices occasionally runs for hours on a confusing case, calling tools hundreds of times and running up large costs. When it finally stops, it returns nothing useful.
>
> The team wants it bounded.

Which two changes should they make? Choose two.

A Enforce a maximum number of turns and a token or spend budget in code

B Ask the agent in the prompt to finish quickly

C Move the agent to the most capable model so it never gets confused

D On hitting a limit, return progress so far with a clear status for a person to pick up

E Remove the agent's tools after the first ten calls without telling it

◆ PAUSE AND DECIDE

Which two changes should they make? Choose two.

A Enforce a maximum number of turns and a token or spend budget in code

Hard stops the agent cannot ignore.

B Ask the agent in the prompt to finish quickly

Guidance, not a limit.

C Move the agent to the most capable model so it never gets confused

No model guarantees termination.

D On hitting a limit, return progress so far with a clear status for a person to pick up

A designed end state.

E Remove the agent's tools after the first ten calls without telling it

Confuses the agent; not a stop.

THE SIGNAL

Two problems: it never stops, and stopping yields nothing. Hard limits, plus a designed end state.

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> A team wants to summarise customer emails into three fields. A tech lead proposes a multi-agent framework with a planner, three specialist agents and a critic. A single call with structured output already meets every criterion on the eval set.
>
> No new requirement is expected.

What should the team do?

A Adopt the multi-agent framework, since it will scale better later

B Add the critic agent only, as a compromise

C Keep the single call with structured output, monitored by the eval set

D Replace the call with an autonomous agent that decides the fields

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What should the team do?

A Adopt the multi-agent framework, since it will scale better later

Complexity for an unstated future.

B Add the critic agent only, as a compromise

Cost with no measured gain.

C Keep the single call with structured output, monitored by the eval set

Simplest solution that passes.

D Replace the call with an autonomous agent that decides the fields

The fields are fixed.

THE SIGNAL

Already passes every criterion: the simplest solution wins. Add complexity only when evals demand it.

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Module 23 in five lines

- **Workflows** follow predefined code paths; **agents** direct their own steps.
- **Start with the augmented LLM;** add structure or autonomy only when needed.
- **Patterns:** chaining, routing, parallelisation, orchestrator-workers, evaluator-optimiser.
- **Agents earn their cost** when steps cannot be predicted and results can be checked.
- **Every agent stops by design:** turn and budget limits, checkpoints, a clear end state.

> COMING UP
>
> Next: orchestrators, subagents and multi-agent hierarchies.